Vineet Vora

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Summary

Engineer who takes AI systems from an ambiguous problem to something deployed and used. MS in Artificial Intelligence at NJIT. Built a natural-language-to-SQL system that lets non-technical users query unfamiliar enterprise databases in plain English, on open-source models only so it runs inside a customer's own environment. Comfortable owning the whole line — scoping, building, standing up the infrastructure, and explaining the result to people who don't write code.

Technical Skills

Languages: Python, Java, TypeScript, JavaScript (ES6+), SQL
AI / ML: PyTorch, HuggingFace Transformers, vLLM, LoRA / PEFT, TensorFlow, Scikit-learn, OpenCV
Data & Infrastructure: Hadoop, MapReduce, HDFS, YARN, AWS EC2, SLURM, Docker, CUDA, PostgreSQL, MongoDB
Web & Delivery: React.js, Node.js, Cloudflare Pages, Git, CI/CD, scoping, Design Thinking (IBM)

Selected Work

Natural Language → SQL over Enterprise Databases (MS Thesis) *Jan 2026 – Present*
Python, vLLM, PyTorch, Qwen3-Coder-30B, SLURM, SQLite

- Turns plain-English business questions into executable SQL against multi-table databases the model has never seen: schema linking, candidate generation, targeted decomposition, self-repair, and consensus voting.
- Reached **70.2% execution accuracy on BIRD** (1,534 questions, 11 real-world databases) using **only open-source models and no proprietary API** — so it runs fully inside a customer's own environment with no data leaving it.
- Served 30B-parameter models on a 2×A100 SLURM cluster with vLLM; built checkpoint/resume, per-question timeouts, and graceful degradation so an 8-hour evaluation never restarts from zero on a single failure.
- Compared four open backbones to isolate what actually moved accuracy — and cut five candidate-selection variants to one after measuring the extra four bought nothing.

Geomagnetic Storm Forecasting from NASA Solar Imagery *Dec 2025 – Present*
github.com/VineetV2/imf-bz-forecasting-surya | Python, PyTorch, LoRA, CUDA, Weights & Biases

- Adapted Surya, a 366M-parameter Vision Transformer pretrained on NASA SDO imagery, to forecast interplanetary magnetic field Bz at +24h/48h/72h — what utilities and satellite operators watch before a geomagnetic storm.
- Fine-tuned on 13-channel AIA EUV and HMI magnetogram inputs (512×512) using LoRA, cutting training cost **3.5**× versus full fine-tuning on a constrained GPU budget.
- Achieved **3.39 nT RMSE**, 96.5% of forecasts rated Excellent/Good, under Leave-One-Flare-Out cross-validation across 57 solar flare events.

Distributed Flight Analytics — 118.9M Records on Hadoop *Nov – Dec 2025*
github.com/VineetV2/hadoop-flight-data-analysis | Java, Hadoop MapReduce, AWS EC2

- Stood up a 6-node Hadoop cluster on AWS EC2 and processed 22 years of US flight data — **118.9M records, 11.66 GB** — with custom MapReduce jobs.
- Achieved **100% data-local map execution** and **2.73× speedup** through horizontal scaling, with sub-linear scaling as data volume grew.

Peon — Local-First Memory System for AI Coding Agents *2026*
github.com/VineetV2/peon-mem | TypeScript, MCP, SQLite

- Built and shipped an MCP server that gives AI coding agents persistent project memory: automatic capture, LLM-based consolidation, hybrid retrieval, and a daily self-audit. Works with any MCP client; used daily as personal engineering tooling.

Publication

Preventing Wildfires in Energy Transmission by Automatic Power Line Defects Detection Using Machine Learning and AI
IEEE Conference, NJIT — December 2025 | presented to a conference audience | ieeexplore.ieee.org/document/11380637

- Detection framework using Mask R-CNN and DeepLabV3+ to find power line defects and vegetation encroachment in UAV imagery, moving utilities from reactive repair to preventive maintenance.

Education

New Jersey Institute of Technology — MS, Artificial Intelligence *Expected Jun 2027*
Newark, NJ | GPA 3.83 | Advanced NLP, Computer Vision, Information Retrieval, ML for Data Science, Big Data

Vellore Institute of Technology – AP — BTech, Computer Science | India *Jun 2025*

Certifications

Enterprise Design Thinking Practitioner (IBM) · Practical GitHub Actions · Learning Docker · Full-Stack Developer Program (Ethnus)